A Three-Level Game-Theoretic Decision-Making Framework for Autonomous Vehicles

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***Abstract*— In this paper, a three-level decision-making framework is developed to generate safe and effective decisions for autonomous vehicles (AVs). A key component in this decision framework is a normal-form game to capture the interactions between the ego vehicle and its surrounding vehicles. The payoffs in the normal-form game are designed to capture both safety reward and the reward gained by obeying (or the price paid by violating) "soft" traffic rules, e.g., first-come-first-go. This game formulation enables the ego to 1) make appropriate decisions considering the payoffs and possible actions of its surrounding vehicles, and 2) take intelligent actions in emergencies that may sacrifice some soft traffic rules to ensure safety. Moreover, we introduce parameters in the payoff matrix to tune the ego vehicle's behavior, e.g., aggressiveness level. A neural network is developed to learn the tuning parameters via supervised learning. In addition, to enable the ego to respond timely to different surrounding vehicles' driving styles, driving style characterization is incorporated into the payoff design for the normal-form game. Simulation studies are conducted to demonstrate the performance of the developed algorithms in two-vehicle intersection-crossing and lane-changing scenarios.**

***Index Terms*— Autonomous driving, decision making, game theory.**

I. Introduction

AUTONOMOUS driving has attracted increasing attention in recent years. The global autonomous vehicle (AV) market is expected to reach $556.67 billion by 2026 with an annual growth of 39.4 percent from 2019 to 2026 [1]. In autonomous driving, developing a safe and effective decision-making solution for AVs is a critical research direction [2]–[4]. AVs are expected to have a lower accident rate compared to human drivers [5]. However, safety is challenging to achieve in mixed manned and unmanned traffic and complex traffic scenarios. Other than safety, it is also important that AVs can operate effectively by considering traffic rules and travel time while they fulfill various tasks such as intersection-crossing and lane-changing. Many studies have been directed to the decision-making of AVs in recent years. In general, these studies can be classified into model-based and data-driven approaches.

Model-based approaches, including simple logical (if-then-else) algorithms, Markov decision process, expert system, multi-agent optimization, and fuzzy logic, are built on predefined action generation models [6]–[11]. In [6], lane-changing behaviors in highways were modeled using "cognitive architectures", from which logical rules were derived for the decision-making of AVs. In [8], decision-making under uncertain drivers' intentions was addressed using the Partially Observable Markov Decision Process (POMDP) approach, with the merging traffic scenario as a case study. In [11], vehicles are designed to collaboratively minimize the total waiting time of all vehicles in intersection-crossing scenarios, using a multi-vehicle optimization approach. In [9], a fuzzy logic -based decision-making approach was developed and tested for the automatic overtaking scenario. These model-based approaches are simple to design, interpret and implement. However, as the models are usually developed for specific traffic scenarios, they do not easily transfer to other scenarios. In addition, to generate decisions for the ego, the models of other vehicles' behaviors are usually required (e.g., assuming that other vehicles are collaborative). However, as the ego cannot control its surrounding vehicles' behaviors, the ego's decision performance (e.g., safety and efficiency) can not be guaranteed when unexpected decisions are enacted by its surrounding vehicles.

Data-driven approaches are becoming popular in devising decision-making solutions for AVs with the rapid advances of machine learning technologies [12]–[15]. In [12], a support vector machine algorithm with Bayesian parameter optimization was applied to the lane-changing scenario. In [13], a deep neural network (DNN) solution was developed with the consideration of general traffic scenarios and real-life road conditions. In [14], a deep reinforcement learning (DRL) algorithm was developed for highway driving. In [15], a Q-learning (QL) algorithm was developed to address the overtaking scenario. In contrast to rule-based approaches, these learning-based approaches take into account the opmatility of decisions. However, they often require high-quality training samples and lacks interpretability, causing trust-related concerns.

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Game theoretic approaches provide us promising mathematical foundations to study interactions of vehicles and allow the design of decision-making for AVs in non-cooperative scenarios by taking the merits of both rule-based and data-driven approaches. In a game formulation, the goal of each vehicle is to optimize its payoff considering the possible actions of other vehicles [2], [2], [16]–[22]. In [19], [20], Stackelberg games are employed, respectively, to solve lane-changing and roundabout-crossing scenarios. In [21], [23], normal-form games and leader-follower games are developed, respectively, to address the intersection-crossing scenarios. In [24], a game-theoretic level-k reasoning method is employed to model the behaviors of human drivers in highway driving scenarios. In [22], two non-cooperative games including Nash equilibrium game and Stackelberg game are proposed and compared to model autonomous racing between two vehicles. In [2], a multi-player game is developed to deal with the coordination of vehicles in highways. The above game-theoretic approaches are all developed for specific traffic scenarios, e.g., lane-changing or intersection-crossing, and lacks the universality to be generalized to diverse traffic scenarios due to the characteristics of some specific game formulations. In addition, in most existing game-theoretic approaches, the payoff functions are designed to capture the safety-requirements only. The traffic rules, although important to the ego's decision-making, are not explicitly incorporated in the game payoff design. Moreover, most of the existing works assume that the ego's surrounding vehicles have the same driving style (e.g., aggressiveness level), which is not practical in real scenarios. The characterization of surrounding vehicles' aggressiveness should also be incorporated into the ego's decision-making. To address the above challenges, in this paper, we propose a novel three-level game theoretic decision-making framework.

The contributions of this paper are fourfold. First, a three-level game-theoretic decision-making framework is developed for general traffic scenarios, where "hard" traffic rules (e.g., not driving wrong way down one-way road) and "soft" rules (e.g., the minimum occupancy requirement of a high-occupancy vehicle (HOV) lane) are separately captured in different levels. Second, we propose a novel payoff design approach, where both safety requirement and soft traffic rules are captured simultaneously. Such payoff design enables the ego to take intelligent actions that may sacrifice soft rules in emergencies to ensure safety. Third, a supervised learning algorithm is developed to learn the tuning parameters in the game payoff matrix. The neural network (NN) structure in the supervised learning is designed analytically according to the game principles, which makes the NN more interpretable compared to the existing data-driven approaches. Fourth, a driving style characterization method is incorporated into the developed framework to enable the ego to respond timely to its surrounding vehicles of various aggressiveness levels.

This paper is organized as follows. Section II reviews the concepts in normal-form games. Section III formulates decision-making problems for AVs in three traffic scenarios and proposes a three-level decision-making framework. Section IV illustrates the design of the payoff matrix in normal-form games with a two-vehicle intersection-crossing scenario as the case study. Section V presents learning methods to tune parameters in payoff matrices. Section VI conducts simulation studies to demonstrate the performance of the designed algorithm, and Section VII concludes the paper.

		Player 2	
		$\boldsymbol{v_1}$	$\boldsymbol{v_2}$
Player 1	$\boldsymbol{u_1}$	a_{11}, b_{11}	a_{12}, b_{12}
	$\boldsymbol{u_2}$	a_{21}, b_{21}	a_{22}, b_{22}

Fig. 1. Two-player normal-form game.

II. Review of Normal-Form Games

In this section, we briefly review some preliminaries in game theory to facilitate the design and analysis in later sections. Formally, a multi-player normal-form game is defined as a tuple $(\mathcal{N}, A, J)$, where $\mathcal{N}$ is the set of N players, $A = A_1 \times A_2 \times \cdots \times A_N$ with A_i the set of actions of player i, and $J = J_1 \times J_2 \times \cdots \times J_N$, with $J_i : A \to \mathbb{R}$ the action-dependent payoff functions of player i [25]. In normal-form games, the goal of each player is to maximize its payoff by selecting an appropriate action that takes into account other players' payoffs associated with their possible actions.

A normal-form game is often represented by a payoff table. Figure 1 shows an example of the two-player normal-form game. In this example, each of the two players, named Player 1 and Player 2, has two possible actions, $A_1 = \{u_1, u_2\}$, and $A_2 = \{v_1, v_2\}$. The payoffs are represented using a_{ij} for Player 1 and b_{ij} for Player 2, where $i, j \in \{1, 2\}$ corresponding to each action in the action space. The top-left cell represents that if Player 1 performs action u_1 and Player 2 performs v_1, then Player 1 receives the payoff a_{11} and Player 2 receives b_{11}. For normal-form games with enlarged action spaces for the players, additional rows and columns need to be added in the payoff table. To consider multi-player ($N > 2$) normal-form games, the dimension of the table needs to be extended from 2 to N.

Let s_i represent the action performed by player i and s_{-i} represent the set of actions performed by all the other players, i.e., $s_{-i} = \{s_1, \cdots, s_{i-1}, s_{i+1}, \cdots, s_N\}$.

Definition 1 (Best Response [25]): The best response of player i, given the action set s_{-i}, is defined as the action s_i^* such that

$$J_i(s_i^*, s_{-i}) \geq J_i(s_i, s_{-i}), \tag{1}$$

for all possible $s_i \in A_i$.

Definition 2 (Nash equilibrium [25]): An N-tuple of actions $\{s_1^*, s_2^*, \ldots, s_N^*\}$ is said to be a Nash equilibrium for an N-player game if for all $i = 1, 2, \cdots, N$,

$$J_i(s_i^*, s_{-i}^*) \geq J_i(s_i, s_{-i}^*) \tag{2}$$

holds for all possible $s_i \in A_i$. That is, if all players perform their best response against each other, a Nash equilibrium is achieved. Nash equilibrium represents a stable status of the

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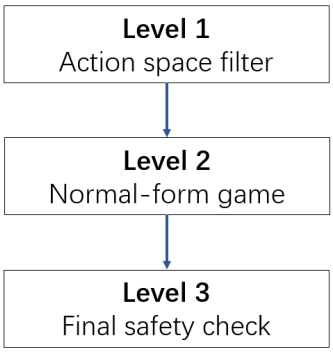

Fig. 2. Three-level decision-making framework.

game in the sense that no player can unilaterally change its decision without decreasing its payoffs. In other words, if a Nash equilibrium is achieved, then no player would have the intention to change its decision.

III. Three-Level Decision-Making Framework for Autonomous Driving

In this section, we first develop a general three-level decision-making framework in Section III-A. We then describe three representative traffic scenarios and illustrate how the framework addresses the specific scenarios in Section III-B. Finally, a repeated play mechanism is introduced in Section III-C.

A. Three-Level Decision-Making Framework

We design a general three-level decision-making framework shown in Figure 2 to generate safe and effective decisions for AVs. Before we detail the functionality of each level, let us first introduce two types of traffic rules, "hard" traffic rules and "soft" traffic rules. Hard traffic rules are the regulations that should never be violated, e.g., not driving wrong way down one-way road. Soft traffic rules refer to the regulations that must be followed under normal driving condition but may be sacrificed at some emergencies to avoid accidents, for example, first-come-first-go in a four-way stop intersection.

The first level of the decision-making framework is an action space filter. Given a scenario, it determines the action space of the decision process and filters out actions that violate hard traffic rules and thus should be removed from the associated action space.

The second level is normal-form game -based decision-making. In this level, the ego plays a normal-form game with its surrounding vehicles, and selects the Nash solution as its decision. In this game-theoretic setting, the ego considers not only its own payoff, but also the possible actions and payoffs of its surrounding vehicles. In other words, the ego is not assumed to be the only vehicle who makes rational decisions. This game setting is consistent with human driving experience. In particular, human drivers observe the behaviors of surrounding vehicles, think about their next steps, and make the best decisions considering both their and their neighbors' payoffs. The payoffs of the normal-form game are desired to capture both safety reward and the reward gained by obeying soft traffic rules.

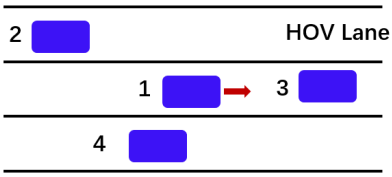

Fig. 3. The lane-changing scenario.

The third level is a final safety check for the selected decision, which is to prevent an unsafe autonomous decision from being implemented. This level is in compliance with the "Responsibility Sensitive Safety (RSS)" model proposed in [5] to prevent the ego from making an accident of its responsibility.

B. Representative Traffic Scenarios

We illustrate in this section how to apply this general framework in three representative traffic scenarios.

1) Scenario 1: Lane-Changing in Highway Traffic: Consider a highway lane-changing scenario shown in Figure 3. Pressed by vehicle 3 in front of it, the ego, labeled with "1" in this scenario desires to determine whether it is appropriate to change to the left lane. To make safe and effective decisions, the ego is desired to optimize a predefined payoff, which depends on multiple factors including traffic rules (i.e., whether it is allowed to drive in a HOV lane), states of vehicle 2 (e.g., positions and velocities), driving styles of vehicle 2 (e.g., aggressive or conservative), and possible actions of vehicle 2 (e.g., accelerate or decelerate).

To address such a decision-making problem, we employ the three-level decision-making framework shown in Figure 2. In the first level, the action space for the ego is defined as $A_1 = \{\text{CLL}, \text{KCL}\}$, where "CLL" represents the action "change to the left lane", and "KCL" represents "keep the current lane". The action space for vehicle 2 is $A_2 = \{\text{RS}, \text{KS}\}$, where "RS" represents "reduce speed" and "KS" represents "keep speed". In the second level, the normal-form game is designed as shown in Figure 4. The payoffs in the game are designed to capture both safety requirement (which can be quantified by, for example, following distance) and soft rules (i.e., the minimum occupancy requirement of the HOV lane). The ego then selects the Nash solution as its decision after playing the normal-form game. In the third level, the ego checks if the selected action from level 2 leads to a pre-defined safe following distance between vehicles 1 and 2. If yes, the ego implements this action. Otherwise, the ego implements a backup safe action, which is "KCL" in this scenario.

By incorporating soft rules in the game payoff, the ego is able to make intelligent decisions in emergencies to ensure safety. For example, if vehicle 4 in Figure 3 performs lane-changing to the left without noticing the ego, then the best decision for the ego would be to change to the HOV lane to avoid collision. Driving in the HOV lane is a violation of soft rule if the minimum occupancy requirement does not

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		Car 2	
		RS	**KS**
Car 1	**CLL**	a_{11}, b_{11}	a_{12}, b_{12}
	KCL	a_{21}, b_{21}	a_{22}, b_{22}

Fig. 4. Normal-form game of lane-changing.

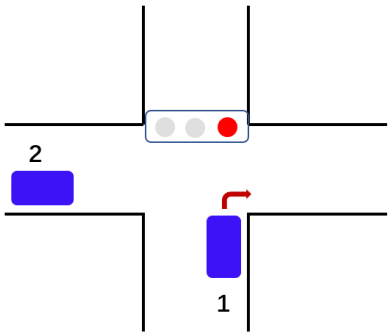

Fig. 5. The right-turning scenario.

		Car 2	
		RS	**KS**
Car 1	**yield**	a_{11}, b_{11}	a_{12}, b_{12}
	go	a_{21}, b_{21}	a_{22}, b_{22}

Fig. 6. Normal-form game of right-turning.

meet. However, such violation is necessary to be captured in a flexible decision framework to avoid accidents in emergencies.

2) Scenario 2: Right-Turning Under a Red Light: Consider an intersection with traffic lights pictured in Figure 5. The ego desires to turn right while its current road has a red light. This decision-making needs consider traffic rules (i.e., whether right-turning is prohibited in this intersection) and vehicle 2's states, driving styles, and possible actions.

The three-level decision framework in this scenario is as follows. In the first level, the hard rule is that a vehicle can turn right only if right-turning under red light is allowed in this intersection. If it is allowed, the action space for the ego is $A_1 = \{\text{go, yield}\}$, where "go" represents the action "turn right", and "yield" represents "wait until vehicle 2 crosses the intersection". The action space for vehicle 2 is $A_2 = \{\text{RS, KS}\}$, where "RS" represents "reduce speed" and "KS" represents "keep speed". In the second level, we define the normal-form game between the ego and vehicle 2 as shown in Figure 6. In the third level, the ego conducts a final safety check with the backup action "yield".

3) Scenario 3: Intersection-Crossing With Stop Signs: Consider a four-way stop sign intersection illustrated in Figure 7. The goal of the ego is to cross the intersection safely and effectively in the presence of vehicle 2. The payoff that the ego desires to optimize captures traffic rules (i.e., first-come-first-go), and vehicle 2's states, driving styles, and possible actions.

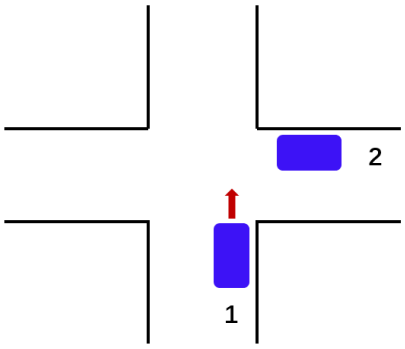

Fig. 7. The intersection-crossing scenario.

		Car 2	
		yield	**go**
Car 1	**yield**	a_{11}, b_{11}	a_{12}, b_{12}
	go	a_{21}, b_{21}	a_{22}, b_{22}

Fig. 8. Normal-form game of intersection-crossing.

In this scenario, the three levels are designed as follows. In the first level, the action space for each vehicle is $A_i = \{\text{go, yield}\}$, where $i = 1, 2$, "go" represents the action "cross the intersection", and "yield" represents "yield to the other vehicle". The hard rule is that a vehicle must come to a complete stop before passing the stop sign. In the second level, the normal form game is shown in Figure 8. The third level conducts a final safety check with the backup action "yield".

We note that the normal-form game in the intersection-crossing scenario, i.e., Figure 8, shares a similar structure with the decision matrix approach in [21]. We summarize the differences of the two approaches as below: 1) The elements in the decision matrix in [21] are designed to capture road priority rules only, while both traffic rules and real-time safety measures are incorporated into our game payoff design, leading to a better safety performance when the surrounding vehicles violate the road rules; 2) The outcome {go, go} is not allowed in [21] when two vehicles are in conflict, while it is admissible in our game when safety is ensured, avoiding unnecessary braking or waiting; and 3) The decision matrix in [21] is designed specifically for intersections with specific layouts, i.e., T-shaped and X-shaped intersections, while our game design is applicable to general intersections and diverse traffic scenarios.

In this paper, we focus our study on two-vehicle scenarios. Note that our framework is applicable to multi-vehicle scenarios by directly employing the pairwise game technique [18]. In pairwise games, the ego plays a 2-player game with each of its surrounding vehicle and selects the safest decision from the game outcomes. Pairwise games may generate safe but inefficient decisions in some multi-vehicle scenarios. This is because the interaction correlations among multiple vehicle

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pairs are neglected in pairwise games, and as such, only "local" interaction between a pair of vehicles is captured. To address this limitation, further study is needed on the generalization of pairwise games to capture the "global" interactions among multiple vehicles, which we leave to the future work.

C. Repeated Play

The above three-level decision-making framework generates safe and effective operations for the ego given the current vehicles' states (e.g., velocities and positions). Note that these state variables are subject to continuous changes with the movement of vehicles. Therefore, the best action for the ego may change frequently. To address that, we let the ego repeat the above decision-making process periodically, e.g. every few milliseconds. The repeated play prepares the ego to re-evaluate its payoffs and decide the best actions according to the most recently acquired data.

Consider, for example, the stop-sign intersection-crossing in Figure 7. The decision-making process provides the ego the best action "go" according to the first-come-first-go rule. However, vehicle 2, instead of yielding to the ego, starts crossing the intersection, which endangers the ego if the ego continues to go. It is thus necessary for the ego to repeat the decision-making process, and at the next time period (after few milliseconds), select the best action as "yield" to vehicle 2 to ensure safety.

In the three-level decision-making framework, the key component is the normal-form game. The payoff matrix design in the normal-form game plays important roles and is illustrated in detail in the next section. From the next section, we choose the two-vehicle intersection-crossing scenario shown in Figure 7 as the representative case study.

IV. Payoff Design

This section describes the design of payoff matrix in the normal-form game shown in Figure 8, for the two-vehicle intersection-crossing scenario in Figure 7. The payoff matrix is critical in determining the ego's behaviors and should be designed to capture both traffic rules and safety requirements. Conceptually, if an action violates the first-come-first-go rule in this stop-sign intersection, then this action should be punished by being assigned a reduced "rule-based payoff" to prevent the ego from selecting this action. Similarly, if an action leads to a dangerous situation, e.g., collision with vehicle 2, then this action should also be punished by a reduced "safety-based payoff". As such, we design the payoffs a_{ij} and b_{ij}, $i, j \in \{1, 2\}$, as

$$a_{ij} = \beta a^s_{ij} + (1-\beta) a^r_{ij}, \tag{3}$$

$$b_{ij} = \beta b^s_{ij} + (1-\beta) b^r_{ij}. \tag{4}$$

Here a^s_{ij} and b^s_{ij} are safety-based payoffs, which reflect how safe the vehicle is by taking the corresponding action. a^r_{ij} and b^r_{ij} are rule-based payoffs, which capture the reward gained by following the soft traffic rules. β is a weighting parameter and satisfies $0 \le \beta \le 1$.

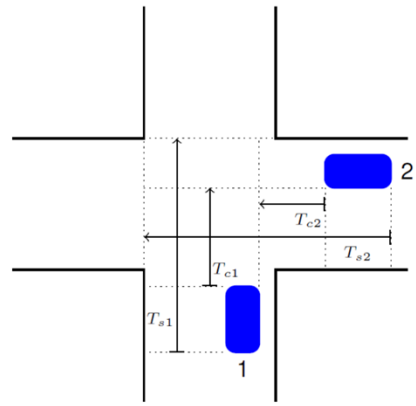

Fig. 9. Decision variables for safety-based payoffs.

Next let us consider the design of the safety-based payoff and rule-based payoff, respectively.

A. Safety-Based Payoff

The safety-based payoff evaluates how safe an action is. We define two decision variables for the safety-based payoff design, time-to-collision (TTC) and time-of-safe-crossing (TOSC).

The TTC for the ego, denoted as T_{c1}, is defined as the time it takes to reach the relative position of vehicle 2 directly in front of the ego as shown in Figure 9.

$$T_{c1} = \frac{d_{12}}{v_1}, \tag{5}$$

where d_{12} is the longitudinal distance between the ego and vehicle 2 along the ego's heading direction, and v_1 is the ego's speed. Similarly, we can define the TTC for vehicle 2 as

$$T_{c2} = \frac{d_{21}}{v_2}, \tag{6}$$

where d_{21} is the latitudinal distance between vehicle 2 and the ego along the direction of vehicle 2's heading, and v_2 is the vehicle 2' speed. If $v_i = 0$, the time-to-collision T_{ci} is set to be a large positive number.

Next we define the TOSC for the ego and vehicle 2. We use T_{s1} to denote the time-of-safe-crossing for the ego. This is the time the ego takes to completely cross the intersection.

$$T_{s1} = \frac{d_1}{v_1}, \tag{7}$$

where d_1 is the longitudinal distance between the ego and the end of the intersection. Similarly, we define the TOSC for vehicle 2 as

$$T_{s2} = \frac{d_2}{v_2}, \tag{8}$$

where d_2 is the latitudinal distance between vehicle 2 and the end of the intersection. If $v_i = 0$, T_{si} is set to be a large positive number.

Next let us define the payoffs a^s_{ij} and b^s_{ij} using the decision variables T_{ci} and T_{si}, where $i, j \in \{1, 2\}$.

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For the ego, its payoff a_{12} if it selects the action "yield" is defined as

$$a_{12}^s = \theta_1 T_{s1} - T_{c2} + \theta_2, \tag{9}$$

where θ_1 and θ_2 are two parameters to tune the ego's behavior. The ego is encouraged to yield if unsafe to cross the intersection, i.e., T_{s1} is large or T_{c2} is small.

Similarly, we define

$$a_{21}^s = a_{22}^s = T_{c2} - \theta_3 T_{s1} + \theta_4, \tag{10}$$

where θ_3 and θ_4 are two tuning parameters. Note that the ego will gain the above two payoffs a_{21}^s and a_{22}^s if it selects the action "go". In other words, if T_{s1} is small and T_{c2} is large, the payoffs a_{21}^s and a_{22}^s should take large values to motivate the ego to select the action "go".

Now let us define a_{11}^s. Note that the outcome "yield/yield" is never desired because it indicates that both vehicles select to yield to each other. Therefore, to prevent the outcome "yield/yield" from being selected, we define a_{11}^s as

$$a_{11}^s = a_{21}^s - \epsilon, \tag{11}$$

where ϵ is any positive constant.

Similarly, the vehicle 2's safety-based payoffs b_{ij}^s are defined as

$$b_{12}^s = b_{22}^s = T_{c1} - \theta_3 T_{s2} + \theta_4, \tag{12}$$
$$b_{21}^s = \theta_1 T_{s2} - T_{c1} + \theta_2, \tag{13}$$
$$b_{11}^s = b_{12}^s - \epsilon. \tag{14}$$

Equation (12) represents that vehicle 2 selects to go only if T_{c1} is small and T_{s2} is large. Equation (13) represents that vehicle 2 selects to yield when T_{s2} is large while T_{c1} is small. b_{11}^s is designed as shown in Equation (14) to prevent the outcome "yield/yield" from being selected as a Nash solution.

B. Rule-Based Payoff

In this subsection, we describe the design of rule-based payoff, i.e., a_{ij}^r and b_{ij}^r. As we are dealing with a four-way stop sign intersection, the rule considered here is first-come-first-go. To capture this rule, we define the relative arriving order r_{12} as our decision variable.

$$r_{12} = \begin{cases} 1, & \text{vehicle 1 arrives earlier than vehicle 2,} \\ -1, & \text{vehicle 2 arrives earlier than vehicle 1,} \\ 0, & \text{vehicles 1 and 2 arrive at the same time.} \end{cases} \tag{15}$$

With the relative arriving order r_{12}, we can define the first-come-first-go -based payoffs a_{ij}^r and b_{ij}^r.

Consider the ego vehicle. Its rule-based payoffs a_{ij}^r are defined as

$$a_{11}^r = a_{12}^r = 0, \tag{16}$$
$$a_{21}^r = a_{22}^r = r_{12}. \tag{17}$$

Comparing Equations (16) and (17) we can see that the ego is motivated to go if r_{12} is positive, i.e., the ego arrives earlier, and is motivated to yield if r_{12} is negative, i.e., the ego arrives later.

Similar payoff designs are applied to vehicle 2.

$$b_{11}^r = b_{21}^r = r_{12}, \tag{18}$$
$$b_{12}^r = b_{22}^r = 0. \tag{19}$$

Combining the safety-based and rule-based payoffs, we can summarize the overall payoffs in the normal-form game shown in Figure 8.

$$a_{11} = \beta(T_{c2} - \theta_3 T_{s1} + \theta_4 - \epsilon), \tag{20}$$
$$a_{12} = \beta(\theta_1 T_{s1} - T_{c2} + \theta_2), \tag{21}$$
$$a_{21} = \beta(T_{c2} - \theta_3 T_{s1} + \theta_4) + (1-\beta) r_{12}, \tag{22}$$
$$a_{22} = \beta(T_{c2} - \theta_3 T_{s1} + \theta_4) + (1-\beta) r_{12}, \tag{23}$$
$$b_{11} = \beta(T_{c1} - \theta_3 T_{s2} + \theta_4 - \epsilon) + (1-\beta) r_{12}, \tag{24}$$
$$b_{12} = \beta(T_{c1} - \theta_3 T_{s2} + \theta_4), \tag{25}$$
$$b_{21} = \beta(\theta_1 T_{s2} - T_{c1} + \theta_2) + (1-\beta) r_{21}, \tag{26}$$
$$b_{22} = \beta(T_{c1} - \theta_3 T_{s2} + \theta_4). \tag{27}$$

Denote $\theta = [\theta_1, \theta_2, \theta_3, \theta_4]^T$. In the above payoff design, we introduce the tuning parameters θ and β to tune the ego's behaviors. If these parameters are selected as constants, then the ego's behaviors are decided uniquely by the decision variables, T_{ci}, T_{si}, and $r_{1,2}$. If these parameters are designed to depend on vehicle 2's driving styles, then the ego is capable of adjusting its decision-making process according to the driving features of others. In the next section, we present a supervised learning approach to incorporate the driving style characterization into the ego vehicle decision making.

V. Learning of Tuning Parameters and Driving Style Characterization

In this section, we describe the learning of tuning parameters θ and β, which are designed as functions of vehicle 2's driving style. Incorporating driving style characterization into the parameter learning enables the ego to respond based on predictive information of different types of driving styles of surrounding vehicles. This design is consistent with human driving experience. If we observe an aggressive vehicle (e.g., speeding up and slowing down frequently and sharply), then we need to drive more conservatively; If we face a conservative vehicle which moves smoothly, then we can act less conservative to improve efficiency. Motivated by this experience, in this section, we first develop a supervised learning mechanism to learn the constant tuning parameters in Section V-A. We then propose a framework to incorporate the learning approach and driving style characterization in Section V-B.

A. Supervised Learning of Tuning Parameters

In this subsection, we design a supervised learning algorithm to learn θ and β. These parameters are regarded as constants in this subsection, and are designed to be functions of vehicle 2's driving style in the next subsection.

We first design a neural network as shown in Figure 10 to facilitate the learning design. This NN has three layers. The input layer consists the decision variables defined in Section IV, i.e., T_{c1}, T_{c2}, T_{s1} T_{s2}, and r_{12}. The second layer

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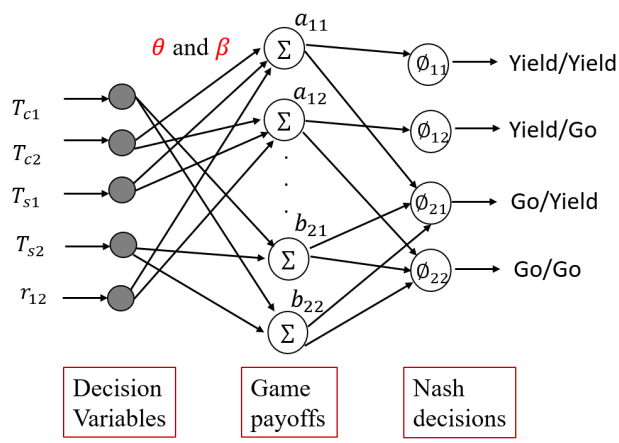

Fig. 10. Incorporation of driving style characterization in game design.

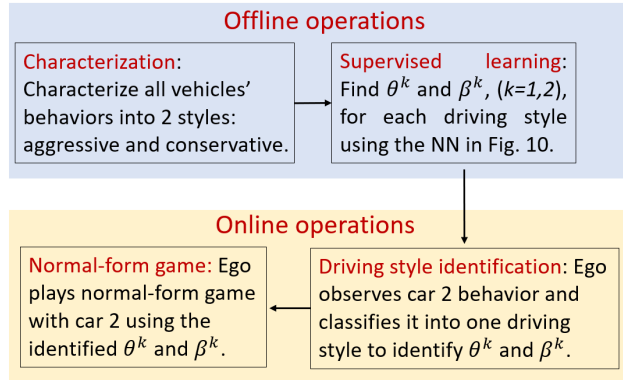

Fig. 11. Three-level decision-making framework.

calculates the game payoffs a_{ij} and b_{ij}. The third layer finds the Nash decision, which is derived by comparing the corresponding payoffs. Here we define the functions ϕ_{ij} in the output layer as

$$\phi_{11} = \tanh(\gamma(a_{11} - a_{12})) + \tanh(\gamma(b_{11} - b_{12})), \quad (28)$$
$$\phi_{12} = \tanh(\gamma(a_{12} - a_{22})) + \tanh(\gamma(b_{12} - b_{11})), \quad (29)$$
$$\phi_{21} = \tanh(\gamma(a_{21} - a_{11})) + \tanh(\gamma(b_{21} - b_{22})), \quad (30)$$
$$\phi_{22} = \tanh(\gamma(a_{22} - a_{12})) + \tanh(\gamma(b_{22} - b_{21})), \quad (31)$$

where γ is a large positive number such that ϕ_{ij} can only take three possible values: -2, 0, and 2. The output neuron that takes the value of 2 is the Nash solution. Consider, for example, Equation (29). $\phi_{12} = 2$ happens only if $a_{12} > a_{22}$ and $b_{12} > b_{11}$, which indicates that the outcome "Yield/Go" is the Nash solution.

With the NN in Figure 10, the values of θ and β are learned by determining the weights of the links between the first two layers (input layer and payoff layer), using supervised learning. Note that the relations between the first two layers are determined by the payoff design (20)-(27), and the relations between the last two layers (payoff layer and output layer) are determined by Equations (28)-(31). As such, only a few unknown parameters, i.e., θ and β, need to be learned by training this NN via supervised learning. After determining the parameters, the payoffs a_{ij} and b_{ij} are determined according to Equations (20)-(27).

With θ and β being constants, the ego always performs the same decision-making process when it faces vehicle 2 of different driving styles. In the next subsections, we design θ and β as functions of vehicle 2's driving style, such that the ego adjusts its decision-making process according to the aggressiveness level of vehicle 2.

B. Normal-Form Games With Driving Style Characterization

In this subsection, we design the tuning parameters θ and β as functions of vehicle 2's driving style. In particular, we first classify all driving behaviors into two styles: aggressive and conservative, and then use the NN in Figure 10 to learn the values of θ and β for each driving style.

The procedures of the normal-form game with driving style characterization are described in Figure 11. The whole procedures are grouped into two main operations: offline operations and online operations. In the offline operations, the ego performs data-driven mapping and learning to determine 1) the driving style characterization criterion, and 2) the values of θ^k and β^k $(k = 1, 2)$ for each driving style k using the NN in Figure 10. In the online operations, the ego 1) characterizes the driving style of vehicle 2, i.e., identify k, according to the characterization criterion, and 2) plays a normal-form game with vehicle 2 with payoffs determined by Equations (20)-(27), θ^k, and β^k.

To label the aggressiveness level of a vehicle, a driving style characterization approach is required. There are many existing works on driving style characterization, and these methods can be generally grouped into the following three categories: 1) mapping the vehicles' trajectory features (e.g., velocities and following distances) to their aggressiveness levels [26]–[29], 2) characterizing the drivers' handling behaviors (e.g., closeness to the handling limits that activate electronic stability control, anti-lock brake system, and traction control system) [30]–[33], and 3) modeling control behaviors (e.g., control gains in a feedback controller that mimics vehicles' dynamics) and mapping them to aggressiveness levels [34], [35]. These driving style characterization techniques can be naturally integrated into our game design by fulfilling the Characterization and Driving style identification steps in Figure 11, to enable the ego to adjust its decision-making process according to other vehicles' aggressiveness levels.

VI. Simulation Studies

In this section, we conduct simulation studies to demonstrate the performance of the proposed decision-making solution in the two-vehicle intersection-crossing scenario.

The vehicles' dynamics is described as follows.

$$\begin{aligned} \dot{x}_i &= v_{x,i}, \\ \dot{y}_i &= v_{y,i}, \\ \dot{v}_{x,i} &= a_{x,i}, \\ \dot{v}_{y,i} &= a_{y,i}, \end{aligned} \quad (32)$$

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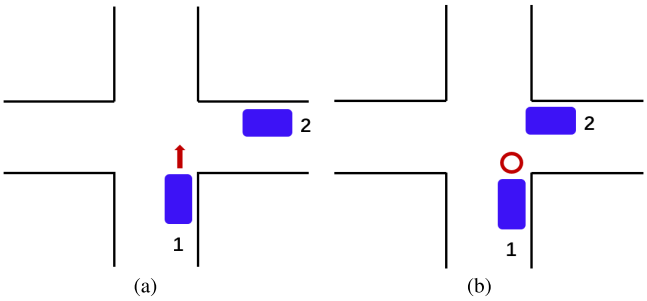

Fig. 12. The ego's decision in Study 1. (a) The ego decides to go as it comes to the intersection first. (b) The ego decides to yield as it comes later.

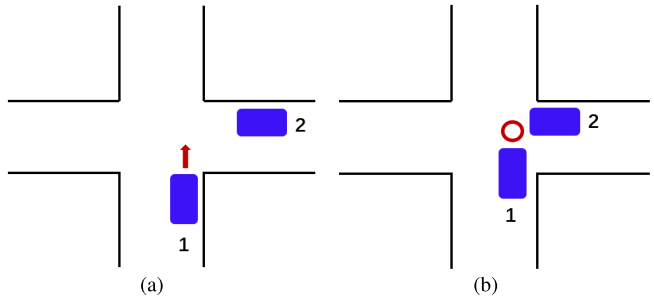

Fig. 14. The ego's decision in Study 2. (a) The ego decides to go as it comes first. (b) After a while, the ego finds vehicle 2 start going, and it then decides to yield to vehicle 2 to ensure safety.

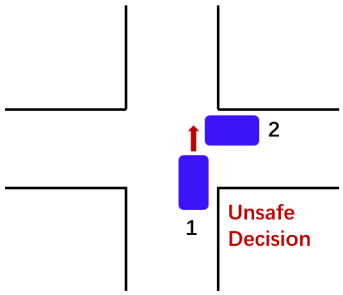

Fig. 13. The ego's decision in Study 1. The ego makes unsafe decision if vehicle 2 violates first-come-first-go rule.

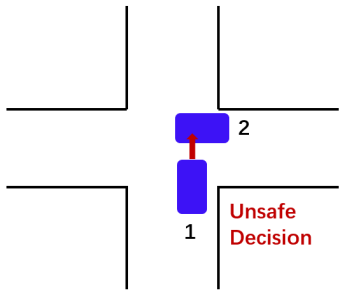

Fig. 15. The ego's decision in Study 2. The tuning parameters θ and β are maliciously-selected.

where $i = 1, 2$ represents the two vehicles, x_i and y_i are the positions of vehicle i along x and y axes, respectively, $v_{x,i}$ and $v_{y,i}$ are the velocities of vehicle i along x and y axes, respectively, and $a_{x,i}$ and $a_{y,i}$ are the accelerations of vehicle i along x and y axes, respectively.

We first simulate the three-level decision-making framework to demonstrate the functionality and necessity of each level in Studies 1-3. In Study 4, we demonstrate the role of driving style characterization when the ego faces vehicle 2 of different driving styles. The effectiveness of the proposed frameworks in other traffic scenarios (e.g., lane-changing in highway traffic) is then demonstrated in Study 5. For all simulation studies, the ego is designed to repeat its decision-making process every 10 milliseconds according to the autonomous driving application needs.

A. Study 1: The First Two Levels With Rule-Based Payoffs Only in Normal-Form Games

In this study, we simulate the performance of the first two levels, where only rule-based payoffs are captured in the normal-form game (i.e., a^r_{ij} and b^r_{ij} in (16)-(19)). We find that the ego performs well in some situations, where vehicle 2 follows the first-come-first-go rule, as shown in Figures 12(a) and 12(b). Here red arrow represents the decision "go" and red circle represents "yield". In the situation shown in Figure 12(a), the ego comes first and thus decides to go after playing the normal-form game. In the situation shown in Figure 12(b), vehicle 2 comes first and the ego decides to yield to give road priority to vehicle 2. Note that both vehicles following first-come-first-go rules can be regarded as a "cooperative" optimization, where the total waiting time is minimized [11]. However, in the situations where vehicle 2 violates the traffic rule (i.e., being non-cooperative), the ego cannot make reasonable decision as shown in Figure 13, where the ego comes first but vehicle 2 does not yield to the ego. In this case, the ego is not able to make safe decision as only rule-based payoffs are considered. The above simulations indicate that rule-based payoffs alone are not sufficient to provide the ego safe decisions.

B. Study 2: The First Two Levels With Both Rule-Based And Safety-Based Payoffs.

Next let us consider the performance of the first two levels, where both rule-based and safety-based payoffs are included as shown in (20)-(27). To provide a comparative study, we select the same initial conditions as the case shown in Figure 13. The decisions of the ego are shown in Figure 14. In an initial decision-making process, the ego decides to go as it comes first. After a while, it finds that vehicle 2 does not follow traffic rules and starts going as well. The ego then decides to yield to vehicle 2 to avoid collision, as shown in Figure 14(b). This simulation indicates that the safety-based payoffs are important to include for the ego to make safe decisions even when vehicle 2 violates the traffic rule or being non-cooperative.

Note that in the above simulations, the tuning parameters θ and β are selected properly via supervised learning. Next we simulate an extreme case where θ and β are maliciously selected as wrong values to mimic a case where the learning algorithm fails to learn θ and β due to the lack of data. Unsurprisingly, the ego fails to make appropriate decisions with

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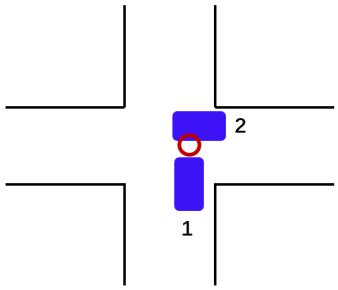

Fig. 16. The ego's decision in Study 3. The tuning parameters θ and β are maliciously-selected.

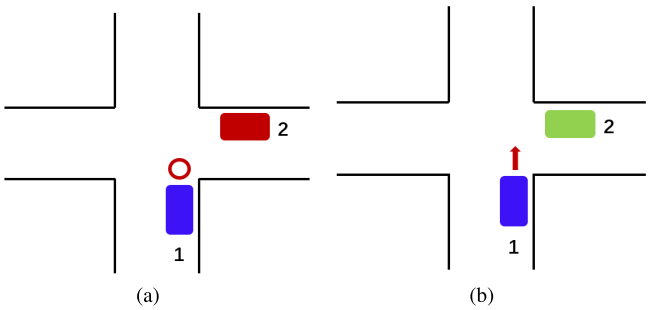

Fig. 17. The ego's decision in Study 4. (a) Aggressive vehicle 2. (b) Conservative vehicle 2.

the maliciously-selected parameters, as shown in Figure 15. This simulation indicates that although the first two levels perform well in normal situations, they may not guarantee safe decisions in unexpected or malicious situations.

C. Study 3: The Complete Three Levels

Next we consider the complete three-level decision-making solution. In the third level, the ego conducts a final safety check. If the action selected from the second level is unsafe (i.e., it leads to small time-to-collision), then the ego takes a backup safe action "yield". With the third level implemented, we re-simulate the case of malicious parameters shown in Figure 15. As shown in Figure 16, the ego is now capable to take a safe action and stop successfully before it collides into vehicle 2.

D. Study 4: Driving Style Characterization

In this study, we simulate the decision-making with driving style characterization illustrated in Section V. We use a simplified rule-based driving style characterization method in the simulation to replace the TDBM as it requires survey-based rating results. The characterization rule is selected as follows. If the average velocity of vehicle 2 exceeds a pre-defined threshold, then we regard it as an aggressive vehicle; otherwise, we regard it as conservative. The decisions of the ego when facing different driving styles of vehicle 2 are shown in Figures 17(a) and 17(b). The red vehicle represents aggressive vehicle 2 and the green one represents conservative vehicle 2. All other settings are selected the same for the two situations. As we can see from Figures 17(a) and 17(b), the ego performs more conservatively when it faces an aggressive vehicle and more assertively when vehicle 2 is conservative. These ego's behaviors are consistent with human driving experiences.

E. Study 5: Simulation In Lane-Changing Scenarios

In this study, we show the effectiveness of the developed framework in lane-changing scenarios in highways, as pictured in Figure 18. It is assumed that the number of people in the ego does not meet the minimum occupancy requirement, i.e., driving in the HOV lane is a violation of soft rule. In the beginning, the ego drives in the middle lane as shown in Figure 18(a). After a while, vehicle 4 begins to perform

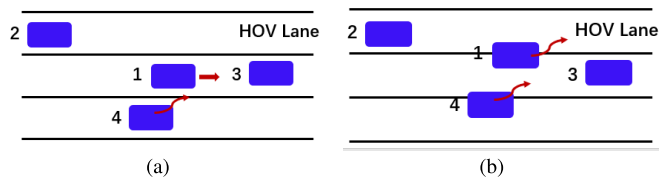

Fig. 18. The ego's performance in a lane-changing scenario. (a) The ego drives in an non-HOV lane without meeting the minimum occupancy requirement. b) Vehicle 4 changes to the ego's lane without noticing the ego, and the ego decides to change to the HOV lane to ensure safety.

left lane-changing without noticing the ego. In this case, by playing a normal-form game with vehicle 4, the ego balances the penalty of violating the soft rule (i.e., driving in the HOV lane) against the cost caused by the collision with vehicle 4, and then decides to change to the HOV lane to ensure safety, as shown in Figure 18(b). This simulation shows that 1) the developed framework is also effective in lane-changing scenarios in highways; and 2) the ego can make intelligent decisions that may sacrifice soft road rules but can ensure safety in emergencies. Note that it is also one of the main principles of RSS [5] that a capable AV should pro-actively avoid a collision. Our solution meets this critical requirement by designing the payoff functions to include both safety requirement and soft rules.

In the above simulation studies, the computational time in one decision-making process (i.e., each play) is less than 1 millisecond, in MATLAB® on a laptop with an Intel Core i7-10750H processor clocked at 2.60 GHz and 16 GB of RAM. In other words, the ego can repeat its decision-making process for every millisecond to respond timely to any changes in its environments. We comment that when generalizing the pairwise game to multi-vehicle scenarios, the computational burden is proportional to the number of surrounding vehicles.

VII. Concluding Remark

In this paper, we developed a three-level decision-making framework to generate safe and effective decisions for autonomous vehicles. To capture the interactions of vehicles, a novel normal-form game was developed, the payoffs of which were designed to capture both safety and soft traffic rules to enable the ego to make intelligent decisions in emergencies. To tune the ego's behavior, we introduced tuning

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parameters in the game payoff. A supervised learning algorithm and a NN were developed to learn the tuning parameters. To enable the ego to respond timely when interacting with surrounding drivers of different driving styles, a driving style characterization was then incorporated into the supervised learning. Simulation studies showed the effectiveness of the developed algorithms. In our future work, we will systematically investigate driving behavior characterization and develop frameworks to generalize the game-theoretic work to multi-vehicle cases.

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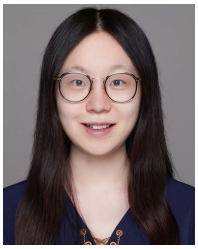

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